Exercise 8.2 SOLUTION

1. Using binomial theorem, expand the following:

i) $\left(\frac{x}{2}-\frac{2}{x^2}\right)^6$

Sol: $\left(\frac{x}{2}-\frac{2}{x^2}\right)^6=\left[\left(\frac{x}{2}\right)+\left(\frac{-2}{x^2}\right)\right]^6$

$=\binom{6}{0}\left(\frac{x}{2}\right)^6\left(\frac{-2}{x^2}\right)^0+\binom{6}{1}\left(\frac{x}{2}\right)^5\left(\frac{-2}{x^2}\right)^1+\binom{6}{2}\left(\frac{x}{2}\right)^4\left(\frac{-2}{x^2}\right)^2+\binom{6}{3}\left(\frac{x}{2}\right)^3\left(\frac{-2}{x^2}\right)^3+\binom{6}{4}\left(\frac{x}{2}\right)^2\left(\frac{-2}{x^2}\right)^4+\binom{6}{5}\left(\frac{x}{2}\right)^1\left(\frac{-2}{x^2}\right)^5+\binom{6}{6}\left(\frac{x}{2}\right)^0\left(\frac{-2}{x^2}\right)^6$

$=(1)\left(\frac{x^6}{64}\right)(1)+(6)\left(\frac{x^5}{32}\right)\left(\frac{-2}{x^2}\right)+(15)\left(\frac{x^4}{16}\right)\left(\frac{4}{x^4}\right)+(20)\left(\frac{x^3}{8}\right)\left(\frac{-8}{x^6}\right)+(15)\left(\frac{x^2}{4}\right)\left(\frac{16}{x^8}\right)+(6)\left(\frac{x}{2}\right)\left(\frac{-32}{x^{10}}\right)+(1)(1)\left(\frac{64}{x^{12}}\right)$

$=\frac{x^6}{64}-\frac{12x^5}{32}+\frac{60x^4}{16x^4}-\frac{160x^3}{8x^6}+\frac{240x^2}{4x^8}-\frac{192x}{2x^{10}}+\frac{64}{x^{12}}=\frac{x^6}{64}-\frac{3x^3}{8}+\frac{15}{4}-\frac{20}{x^3}+\frac{60}{x^6}-\frac{96}{x^9}+\frac{64}{x^{12}}$

ii) $\left(2a-\frac{x}{a}\right)^7$

Sol: $\left(2a-\frac{x}{a}\right)^7$

$\left(2a-\frac{x^2}{a}\right)^7=\left[2a+\left(-\frac{x^2}{a}\right)\right]^7=\binom{7}{0}(2a)^7\left(\frac{-x^2}{a}\right)^0+\binom{7}{1}(2a)^6\left(\frac{-x^2}{a}\right)^1+\binom{7}{2}(2a)^5\left(\frac{-x^2}{a}\right)^2+\binom{7}{3}(2a)^4\left(\frac{-x^2}{a}\right)^3$

$+\binom{7}{4}(2a)^3\left(\frac{-x^2}{a}\right)^4+\binom{7}{5}(2a)^2\left(\frac{-x^2}{a}\right)^5+\binom{7}{6}(2a)^1\left(\frac{-x^2}{a}\right)^6+\binom{7}{7}(2a)^0\left(\frac{-x^2}{a}\right)^7$

$=(1)(128a^7)(1)+(7)(64a^6)\left(-\frac{x^2}{a}\right)+(21)(32a^5)\left(\frac{x^4}{a^2}\right)+(35)(16a^4)\left(-\frac{x^6}{a^3}\right)+(35)(8a^3)\left(\frac{x^8}{a^4}\right)$

$+(21)(4a^2)\left(-\frac{x^{10}}{a^5}\right)+(7)(2a)\left(\frac{x^{12}}{a^6}\right)+(1)(1)\left(-\frac{x^{14}}{a^7}\right)$

$=128a^7-\frac{448a^6x^2}{a}+\frac{672a^5x^4}{a^2}-\frac{560a^4x^6}{a^3}+\frac{280a^3x^8}{a^4}-\frac{84a^2x^{10}}{a^5}+\frac{14ax^{12}}{a^6}-\frac{x^{14}}{a^7}$

$=128a^7-448a^5x^2+672a^3x^4-560ax^6+\frac{280x^8}{a^4}-\frac{84x^{10}}{a^3}+\frac{14x^{12}}{a^5}-\frac{x^{14}}{a^7}$

iii) $\left(\sqrt{\frac{a}{x}}-\sqrt{\frac{x}{a}}\right)^6$

Sol: $\left(\sqrt{\frac{a}{x}}-\sqrt{\frac{x}{a}}\right)^6=\left[\sqrt{\frac{a}{x}}+\left(\sqrt{\frac{-x}{a}}\right)\right]^6$

$=\binom{6}{0}\left(\sqrt{\frac{a}{x}}\right)^6\left(-\sqrt{\frac{x}{a}}\right)^0+\binom{6}{1}\left(\sqrt{\frac{a}{x}}\right)^5\left(-\sqrt{\frac{x}{a}}\right)^1+\binom{6}{2}\left(\sqrt{\frac{a}{x}}\right)^4\left(-\sqrt{\frac{x}{a}}\right)^2+\binom{6}{3}\left(\sqrt{\frac{a}{x}}\right)^3\left(-\sqrt{\frac{x}{a}}\right)^3+\binom{6}{4}\left(\sqrt{\frac{a}{x}}\right)^2\left(-\sqrt{\frac{x}{a}}\right)^4+\binom{6}{5}\left(\sqrt{\frac{a}{x}}\right)^1\left(-\sqrt{\frac{x}{a}}\right)^5+\binom{6}{6}\left(\sqrt{\frac{a}{x}}\right)^0\left(-\sqrt{\frac{x}{a}}\right)^6$

$=(1)\left(\frac{a^{\frac{6}{2}}}{x^{\frac{6}{2}}}\right)(1)+(6)\left(\frac{a^{\frac{5}{2}}}{x^{\frac{5}{2}}}\right)\left(\frac{-x^{\frac{1}{2}}}{a^{\frac{1}{2}}}\right)+(15)\left(\frac{a^{\frac{4}{2}}}{x^{\frac{4}{2}}}\right)\left(\frac{x}{a}\right)+(20)\left(\frac{a^{\frac{3}{2}}}{x^{\frac{3}{2}}}\right)\left(\frac{-x^{\frac{3}{2}}}{a^{\frac{3}{2}}}\right)+(15)\left(\frac{a^{\frac{2}{2}}}{x^{\frac{2}{2}}}\right)\left(\frac{x^{\frac{4}{2}}}{a^{\frac{4}{2}}}\right)+(6)\left(\frac{a^{\frac{1}{2}}}{x^{\frac{1}{2}}}\right)\left(\frac{-x^{\frac{5}{2}}}{a^{\frac{5}{2}}}\right)+(1)(1)\left(\frac{x^{\frac{6}{2}}}{a^{\frac{6}{2}}}\right)$

$=\frac{a^3}{x^3}-\frac{6x^{\frac{5}{2}-\frac{1}{2}}}{a^{\frac{5}{2}-\frac{1}{2}}}+\frac{15a^2x}{x^2a}-20+\frac{15ax^2}{ax^2}-\frac{6x^{\frac{5}{2}-\frac{1}{2}}}{a^{\frac{5}{2}-\frac{1}{2}}}+\frac{x^3}{a^3}=\frac{a^3}{x^3}-\frac{6a^2}{x^2}+\frac{15a}{x}-20+\frac{15}{a}-\frac{6x^2}{a^9}+\frac{x^3}{a^3}$

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2. Calculate the following by means of binomial theorem:

i) $(0.97)^3$

Sol: $(0.97)^3 = (1-0.03)^3 = [1+(-0.03)]^3$

$$= \binom{3}{0}(1)^3(-0.03)^0 + \binom{3}{1}(1)^2(-0.03)^1 + \binom{3}{2}(1)^1(-0.03)^2 + \binom{3}{3}(1)^0(-0.03)^3$$

$$= (1)(1)(1) + (3)(1)(-0.03) + (3)(1)(0.0009) + (-0.000027)$$

$$= 1 - 0.09 + 0.0027 - 0.000027 = 0.912673$$

نوٹ

سوال میں بریکٹ کے اندر جو نمبر دیا گیا ہے اس کے قریب ترین مکمل نمبر میں جمع یا منفی کرنے سے یہ نمبر آتا ہے۔ اس طرح دو نمبر آئیں گے اور Binomial لگا کر حل کریں

ii) $(2.02)^4$

Sol: $(2.02)^4 = (2+0.02)^4$

$$= \binom{4}{0}(2)^4(0.02)^0 + \binom{4}{1}(2)^3(0.02)^1 + \binom{4}{2}(2)^2(0.02)^2 + \binom{4}{3}(2)^1(0.02)^3 + \binom{4}{4}(2)^0(0.02)^4$$

$$= (1)(16)(1) + (4)(8)(0.02) + (6)(4)(0.0004) + (4)(2)(0.000008) + (1)(1)(0.00000016)$$

$$= 16.64966416$$

iii) $(9.98)^4$

Sol: $(9.98)^4 = (10-0.02)^4 = [10+(-0.02)]^4$

$$= \binom{4}{0}(10)^4(-0.02)^0 + \binom{4}{1}(10)^3(-0.02)^1 + \binom{4}{2}(10)^2(-0.02)^2 + \binom{4}{3}(10)^1(-0.02)^3 + \binom{4}{4}(10)^0(-0.02)^4$$

$$= (1)(10000)(1) + (4)(1000)(-0.02) + (6)(100)(0.0004) + (4)(10)(0.000008) + (1)(1)(0.00000016)$$

$$= 9920.23968$$

iv) $(2.1)^5$

Sol: $(2.1)^5 = (2+0.1)^5$

$$= \binom{5}{0}(2)^5(0.1)^0 + \binom{5}{1}(2)^4(0.1)^1 + \binom{5}{2}(2)^3(0.1)^2 + \binom{5}{3}(2)^2(0.1)^3 + \binom{5}{4}(2)^1(0.1)^4 + \binom{5}{5}(2)^0(0.1)^5$$

$$= (1)(32)(1) + (5)(16)(0.1) + (10)(8)(0.01) + +(10)(4)(0.001) + (5)(2)(0.0001) + (1)(1)(0.00001)$$

$$= 32 + 8 + 0.8 + 0.04 + 0.001 + 0.00001 = 40.84101$$

3. Expand and simplify the following:

i) $\left(a+\sqrt{2}x\right)^4 + \left(a-\sqrt{2}x\right)^4$

Sol: $\left(a+\sqrt{2}x\right)^4 + \left(a-\sqrt{2}x\right)^4$

Take $\left(a+\sqrt{2}x\right)^4$

$$= \binom{4}{0}(a)^4\left(\sqrt{2}x\right)^0 + \binom{4}{1}(a)^3\left(\sqrt{2}x\right)^1 + \binom{4}{2}(a)^2\left(\sqrt{2}x\right)^2 + \binom{4}{3}(a)^1\left(\sqrt{2}x\right)^3 + \binom{4}{4}(a)^0\left(\sqrt{2}x\right)^4$$

$$= (1)(a^4)(1) + (4)(a^3)\left(\sqrt{2}x\right) + (6)(a^2)(2x^2) + (4)(a)\left(2\sqrt{2}x^3\right) + (1)(1)(4x^4) = a^4 + 4\sqrt{2}a^3 + 12a^2x^2 + 8\sqrt{2}ax^3 + 4x^4$$

So, $\left(a+\sqrt{2}x\right)^4 = a^4 + 4\sqrt{2}a^3x + 12a^2x^2 + 8\sqrt{2}ax^3 + 4x^4 \quad (i)$

Similarly $\left(a-\sqrt{2}x\right)^4 = a^4 - 4\sqrt{2}ax^3 + 12a^2x^2 - 8\sqrt{2}ax^3 + 4x^4 \quad (ii)$

Adding (i) and (ii), we get

$\left(a+\sqrt{2}x\right)^4 + \left(a-\sqrt{2}x\right)^4 = 2a^4 + 24a^2x^2 + 8x^4$ Which is required.

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$\left(2+\sqrt{3}\right)^5+\left(2-\sqrt{3}\right)^5$

Take $\left(2+\sqrt{3}\right)^5$

$$=\binom{5}{0}(2)^5\left(\sqrt{3}\right)^0+\binom{5}{1}(2)^4\left(\sqrt{3}\right)^1+\binom{5}{2}(2)^3\left(\sqrt{3}\right)^2+\binom{5}{3}(2)^2\left(\sqrt{3}\right)^1+\binom{5}{4}(2)^1\left(\sqrt{3}\right)^4+\binom{5}{5}(2)^0\left(\sqrt{3}\right)^5$$

$$=(1)(32)(1)+(5)(16)\left(\sqrt{3}\right)+(10)(8)(3)+(10)(4)\left(3\sqrt{3}\right)+(5)(2)(9)+(1)(1)\left(9\sqrt{3}\right)$$

$$=32+80\sqrt{3}+240+120\sqrt{3}+90+9\sqrt{3} \qquad \therefore \left(\sqrt{3}\right)^5=\left(\sqrt{3}\right)\left(\sqrt{3}\right)^4=9\sqrt{3}$$

So, $\left(2+\sqrt{3}\right)^5=32+80\sqrt{3}+240+120\sqrt{3}+90+9\sqrt{3} \qquad (i)$

Similarly, $\left(2-\sqrt{3}\right)^5=32-80\sqrt{3}+240-120\sqrt{3}+90-9\sqrt{3} \qquad (ii)$

Adding (i) and (ii), we get

$$\left(2+\sqrt{3}\right)^5+\left(2-\sqrt{3}\right)^5=64+480+180=724$$

Expand the following in ascending power of x:

i) $\left(2+x-x^2\right)^4$

Sol: $\left(2+x-x^2\right)^4=\left[(2+x)-\left(x^2\right)\right]^4=\left[(2+x)+\left(-x^2\right)\right]^4$

نوٹ

اس سوال میں تین نمبر دیے گئے ہیں تو x^2 کے علاوہ باقی دونوں نمبرز کو ایک نمبر فرض کریں اور پھر دو نمبرز پر Binomial لگائیں۔

$$=\binom{4}{0}(2+x)^4\left(-x^2\right)^0+\binom{4}{1}(2+x)^3\left(-x^2\right)^1+\binom{4}{2}(2+x)^2\left(-x^2\right)^2+\binom{4}{3}(2+x)^1\left(-x^2\right)^3+\binom{4}{4}(2+x)^0\left(-x^2\right)^4$$

$$=(1)(2+x)^4(1)+(4)\left(8+12x+6x^2+x^3\right)\left(-x^2\right)+(6)\left(4+4x+x^2\right)\left(x^4\right)+(4)(2+x)\left(-x^6\right)+(1)(1)\left(x^8\right)$$

$$=16+32x+24x^2+8x^3+x^4-32x^2-48x^3-24x^4-4x^5+24x^4+24x^5+6x^6-8x^6-4x^7+x^8$$

$$=16+32x+24x^2-32x^2+8x^3-24x^3+x^4-24x^4+24x^4-4x^5+24x^5+6x^6-8x^6-4x^7+x^8$$

$$=16+32x-8x^2-40x^3+x^4+20x^5-2x^6-4x^7+x^8$$

ii) $\left(1-x+x^2\right)^4$

Sol: $\left(1-x+x^2\right)^4=\left[(1-x)+\left(x^2\right)\right]^4$

$$=\binom{4}{0}(1-x)^4\left(x^2\right)^0+\binom{4}{1}(1-x)^3\left(x^2\right)^1+\binom{4}{2}(1-x)^2\left(x^2\right)^2+\binom{4}{3}(1-x)^1\left(x^2\right)^3+\binom{4}{4}(1-x)^0\left(x^2\right)^4$$

$$=(1)\left(1-4x+6x^2-4x^3+x^4\right)(1)+(4)\left(1-3x+3x^2-x^3\right)\left(x^2\right)+(6)\left(1-2x+x^2\right)\left(x^4\right)+(4)(1-x)\left(x^6\right)+(1)(1)\left(x^8\right)$$

$$=1-4x+6x^2-4x^3+x^4+4x^2-12x^3+12x^4-4x^5+6x^4-12x^5+6x^6+4x^6-4x^7+x^8$$

$$=1-4x+6x^2-4x^3+x^4+4x^2-12x^3+12x^4-4x^5+6x^4-12x^5+6x^6+4x^6-4x^7+x^8$$

$$=1-4x+10x^2-16x^3+19x^4-16x^5+10x^6-4x^7+x^8$$

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5. Find the term involving:

i) x^4 in the expansion of $(3-2x)^7$.

Sol: x^4 in the expansion of $(3-2x)^7$

Given $(3-2x)^7 = [3+(-2x)]^7$

Its general term is $\therefore \quad T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Putting the values, we get

$$T_{r+1} = \binom{7}{r} (3)^{7-r} (-2x)^r$$

$$T_{r+1} = \binom{7}{r} (3)^{7-r} (-2)^r x^r \qquad (i)$$

For the term involving x^4, put $r = 4$

$$\therefore T_{4+1} = \binom{7}{4} (3)^{7-4} (-2)^4 x^4$$

$$T_5 = 35(3)^3 (2)^4 x^4$$

$$T_5 = 35(27)(16)x^4$$

$$T_5 = 15120x^4 \quad \text{Which is required.}$$

ii) x^{-2} in the expansion of $\left(x - \frac{2}{x^2}\right)^{13}$

Sol: x^{-2} in the expansion of $\left(x - \frac{2}{x^2}\right)^{13}$

Given $\left(x - \frac{2}{x^2}\right)^{13} = \left[x + \left(-\frac{2}{x^2}\right)\right]^{13}$

Its general term is $\therefore \quad T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Putting the values, we get

$$T_{r+1} = \binom{13}{r} (x)^{13-r} \left(-\frac{2}{x^2}\right)^r$$

$$T_{r+1} = \binom{13}{r} x^{13-r} (-2)^r \cdot \left(-\frac{1}{x^{2r}}\right)$$

$$T_{r+1} = \binom{13}{r} x^{13-r-2r} (-2)^r$$

$$T_{r+1} = \binom{13}{r} x^{13-r-2r} (-2)^r \qquad (i)$$

For the term involving x^{-2}, put $13 - 3r = -2$

$\Rightarrow 13 + 2 = 3r \Rightarrow 15 = 3r \Rightarrow r = 5$ in (i)

So $T_{5+1} = \binom{13}{5} x^{13-3(5)} \cdot (-2)^5$

$$T_6 = (1287)x^{-2}(-32)$$

$$T_6 = -41184x^{-2} \quad \text{Which is required.}$$

iii) a^4 in the expansion of $\left(\frac{2}{x} - a\right)^9$

Sol: a^4 in the expansion of $\left(\frac{2}{x} - a\right)^9$

Given $\left(\frac{2}{x} - a\right)^9 = \left[\frac{2}{x} + (-a)\right]^9$

Its general term is $\therefore \quad T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Putting the values, we get

$$T_{r+1} = \binom{9}{r} \left(\frac{2}{x}\right)^{9-r} (-a)^9$$

$$T_{r+1} = \binom{9}{r} \frac{2}{x^{9-r}}^{9-r} \cdot (-1)^r (a)^r \qquad (i)$$

For the term involving a^4, put $r = 4$ in (i)

$$T_{4+1} = \binom{9}{4} \frac{2}{x^{9-4}}^{9-4} \cdot (-1)^4 (a)^4$$

$$T_5 = 126 \frac{2^5}{x^5} . a^4 = \frac{4032a^4}{x^5} \quad \text{Which is required.}$$

iv) y^3 in the expansion of $\left(x - \sqrt{y}\right)^{11}$

Sol: y^3 in the expansion of $\left(x - \sqrt{y}\right)^{11}$

Given $\left(x - \sqrt{y}\right)^{11} = \left[x + \left(-\sqrt{y}\right)\right]^{11}$

Its general term is $\therefore \quad T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Putting the values, we get

$$T_{r+1} = \binom{11}{r} (x)^{11-r} \left(-\sqrt{y}\right)^r$$

$$T_{r+1} = \binom{11}{r} x^{11-r} \cdot (-1)^r (y)^{\frac{r}{2}} \qquad (i)$$

For the required term involving y^3, put $\frac{r}{2} = 3$ in

(i) $\Rightarrow r = 6$

$$\therefore T_{6+1} = \binom{11}{6} x^{11-6} \cdot (-1)^6 (y)^{\frac{6}{2}}$$

$$T_7 = (462)(x)^5 (1) y^3$$

$$T_7 = 462x^5 y^3 \quad \text{Which is required.}$$

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6. **Find the coefficient of:**

i) **x^5 in the expansion of $\left(x^2-\frac{3}{2x}\right)^{10}$**

Sol: x^5 in the expansion of $\left(x^2-\frac{3}{2x}\right)^{10}$

Given $\left(x^2-\frac{3}{2x}\right)^{10}=\left[x^2+\left(\frac{-3}{2x}\right)\right]^{10}$

Its general term is $\therefore \quad T_{r+1}=\binom{n}{r}a^{n-r}b^r$

Putting the values, we get

$$T_{r+1}=\binom{10}{r}(x^2)^{10-r}\left(\frac{-3}{2x}\right)^r$$

$$T_{r+1}=\binom{10}{r}x^{20-2r}\left(\frac{-3}{2}\right)^r\cdot\left(\frac{1}{x^r}\right)$$

$$T_{r+1}=\binom{10}{r}x^{20-2r-r}\left(\frac{-3}{2}\right)^r \qquad (i)$$

For the term involving x^5,
Put $20-3r=5 \Rightarrow 20-5=3r \Rightarrow 15=3r \Rightarrow r=5$
In equation (i), we get

$$\therefore T_{5+1}=\binom{10}{5}x^{20-15}\left(\frac{-3}{2}\right)^5$$

$$T_6=(252)x^5\left(\frac{-243}{32}\right)$$

$$T_6=\frac{-61236x^5}{32}=-\frac{15309}{8}x^5$$

So, coefficient of x^5 is $-\frac{15309}{8}$

ii) **x^n in the expansion of $\left(x^2-\frac{1}{x}\right)^{2n}$**

Sol: x^5 in the expansion of $\left(x^2-\frac{1}{x}\right)^{2n}$

Given $\left(x^2-\frac{1}{2x}\right)^{2n}=\left[x^2+\left(-\frac{1}{x}\right)\right]^{2n}$

Its general term is $\therefore \quad T_{r+1}=\binom{n}{r}a^{n-r}b^r$

Putting the values, we get

$$T_{r+1}=\binom{2n}{r}(x^2)^{2n-r}\cdot\left(-\frac{1}{x}\right)^r$$

$$T_{r+1}=\binom{2n}{r}x^{4n-2r}(-1)^r\cdot\frac{1}{x^r}$$

$$T_{r+1}=\binom{2n}{r}x^{4n-2r-r}(-1)^r$$

$$T_{r+1}=\binom{2n}{r}x^{4n-3r}(-1)^r \qquad (i)$$

For the term involving x^n, put $4n-n=3r$
$\Rightarrow 4n-n=3r \Rightarrow 3n=3r \Rightarrow r=n$ in (i),
we get

$$T_{n+1}=\binom{2n}{n}x^{4n-3n}.(-1)^n=\frac{(2n)!}{n!n!}.(-1)^n x^n$$

$$T_{n+1}=\frac{(2n)!}{n!(2n-n)!}.(-1)^n.x^n=\frac{(2n)!}{n!n!}.(-1)^n.x^n$$

$$T_{n+1}=\frac{(2n)!}{(n!)^2}.(-1)^n.x^n$$

So, coefficient of x^n is $\frac{(2n)!}{(n!)^2}.(-1)^n$

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7. **Find 6th term in the expansion of $\left(x^2-\frac{3}{2x}\right)^{10}$**

Sol: 6th term in the expansion of $\left(x^2-\frac{3}{2x}\right)^{10}$

6th term $T_6=?$

Given $\left(x^2-\frac{3}{2x}\right)^{10}=\left(x^2+\left(\frac{-3}{2x}\right)\right)^{10}$

Its general term is $\therefore \quad T_{r+1}=\binom{n}{r}a^{n-r}b^r$

Putting the values, we get

$$T_{r+1}=\binom{10}{r}(x^2)^{10-r}\left(\frac{-3}{2x}\right)^r$$

$$T_{r+1}=\binom{10}{r}(x)^{20-2r}\left(\frac{-3}{2}\right)^r\cdot\frac{1}{x^r}$$

$$T_{r+1}=\binom{10}{r}(x)^{20-2r-r}\left(\frac{-3}{2}\right)^r$$

$$T_{r+1}=\binom{10}{r}(x)^{20-3r}\left(\frac{-3}{2}\right)^r$$

For 6th term, put $r=5$

$$T_{5+1}=\binom{10}{5}(x)^{20-3(5)}\left(\frac{-3}{2}\right)^5$$

$$T_{5+1}=252x^5.\left(\frac{-243}{32}\right)^5=\frac{-15309}{8}x^5$$

Which is required.

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8. Find the term independent of x in the following expansions.

i) $\left(x-\frac{2}{x}\right)^{10}$

Sol: $\left(x-\frac{2}{x}\right)^{10}$ Term independent of x=?

Given $\left(x-\frac{2}{x}\right)^{10}=\left[x+\left(\frac{-2}{x}\right)\right]^{10}$

Its general term is $\therefore \quad T_{r+1}=\binom{n}{r}a^{n-r}b^{r}$

Putting the values, we get

$$T_{r+1}=\binom{10}{r}(x)^{10-r}\left(\frac{-2}{x}\right)^{r}$$

$$T_{r+1}=\binom{10}{r}(x)^{10-r}(-2)^{r}.\frac{1}{x^{r}}$$

$$T_{r+1}=\binom{10}{r}x^{10-r-r}(-2)^{r}=\binom{10}{r}x^{10-2r}.(-2)^{r} \quad (i)$$

For the term independent of x, put $10-2r=0$

$\Rightarrow 10=2r \Rightarrow, r=5$

$$T_{5+1}=\binom{10}{5}x^{10-2(5)}(-2)^{5}$$

$$T_{6}=252x^{0}(-32)=-252\times1\times32=-8064$$

So, the term independent of x is -8604.

ii) $\left(\sqrt{x}+\frac{1}{2x^{2}}\right)^{10}$

Sol: $\left(\sqrt{x}+\frac{1}{2x^{2}}\right)^{10}$ Term independent of x=?

Its general term is $\therefore \quad T_{r+1}=\binom{n}{r}a^{n-r}b^{r}$

Putting the values, we get

$$T_{r+1}=\binom{10}{r}\left(\sqrt{x}\right)^{10-r}\left(\frac{1}{2x^{2}}\right)^{r}$$

$$T_{r+1}=\binom{10}{r}x^{\frac{1}{2}(10-r)}\frac{1}{2^{r}}.\frac{1}{x^{2r}}$$

$$T_{r+1}=\binom{10}{r}x^{\frac{10}{2}-\frac{r}{2}-2r}.\frac{1}{2^{r}}$$

$$T_{r+1}=\binom{10}{r}x^{5-\frac{5}{2}r}.\frac{1}{2^{r}} \quad (i)$$

For the term independent of x

Put $5-\frac{5}{2}r=0 \quad \Rightarrow 5=\frac{5}{2}r \Rightarrow 10=5r$

$\Rightarrow r=2$ in eq (i), we get

$$T_{2+1}=\binom{10}{2}x^{5-\frac{5}{2}(2)}.\frac{1}{2^{2}}$$

$$T_{3}=45.x^{0}.\frac{1}{4}=\frac{45\times1}{4}=\frac{45}{4}$$

So, the term independent of x is $\frac{45}{4}$.

iii) $\left(1+x^{2}\right)^{3}\left(1+\frac{1}{x^{2}}\right)^{4}$

$\left(1+x^{2}\right)^{3}\left(1+\frac{1}{x^{2}}\right)^{4}$ Term independent of x=?

Given $\left(1+x^{2}\right)^{3}\left(1+\frac{1}{x^{2}}\right)^{4}$

$$=\left(1+x^{2}\right)^{3}\left(1+\frac{1}{x^{2}}\right)^{4}=\frac{\left(1+x^{2}\right)^{3}\left(1+x^{2}\right)^{4}}{x^{8}}$$

$$=\frac{\left(1+x^{2}\right)^{3+4}}{x^{8}}=\frac{\left(1+x^{2}\right)^{7}}{x^{8}}=\frac{\left(1+x^{2}\right)^{7}}{x^{7\times\frac{8}{7}}}=\left(\frac{1+x^{2}}{x^{\frac{8}{7}}}\right)^{7}$$

$$=\left(\frac{1}{x^{\frac{8}{7}}}+x^{2-\frac{8}{7}}\right)^{7}=\left(\frac{1}{x^{\frac{8}{7}}}+x^{\frac{6}{7}}\right)^{7}=\left(x^{\frac{6}{7}}+\frac{1}{x^{\frac{8}{7}}}\right)^{7}$$

Its general term is $T_{r+1}=\binom{n}{r}a^{n-r}.b^{r}$

Putting the values

$$T_{r+1}=\binom{7}{r}\left(x^{\frac{6}{7}}\right)^{7-r}.\left(\frac{1}{x^{\frac{8}{7}}}\right)^{r}$$

$$T_{r+1}=\binom{7}{r}x^{6-\frac{6r}{7}}.\frac{1}{x^{\frac{8}{7}r}}$$

$$T_{r+1}=\binom{7}{r}x^{6-\frac{6r}{7}-\frac{8r}{7}}$$

$$T_{r+1}=\binom{7}{r}x^{6-\frac{14r}{7}}=\binom{7}{r}x^{6-2r} \quad (i)$$

For the term independent of x,

Put $6-2r=0 \quad \Rightarrow 6=2r \Rightarrow r=3$ in (i)

$$T_{3+1}=\binom{7}{r}x^{6-2(3)}=35x^{0}=35$$

So, the term independent of x is 35.

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نوٹ

سوال میں جو بھی power دی گئی ہے۔ تو اس میں ایک جمع کرنے سے ٹوٹل ٹرم آئیں گی اگر n=12 ہے تو اس کے جواب میں 13 ٹرم ہیں جس کی مڈل ٹرم 7 ہے اسے فارمولا سے بھی معلوم کیا جا سکتا ہے۔

9. Determine the middle term in the following expansions:

i) $\left(\frac{1}{x}-\frac{x^2}{2}\right)^{12}$

Sol: $\left(\frac{1}{x}-\frac{x^2}{2}\right)^{12}$ Middle term=?

Given $\left(\frac{1}{x}-\frac{x^2}{2}\right)^{12}=\left[\frac{1}{x}+\left(\frac{-x^2}{2}\right)\right]^{12}$

Power $n=12$ is even

$\therefore$ Middle term $=\left(\frac{n}{2}+1\right)^{th}$ $term$

$=\left(\frac{12}{2}+1\right)^{th}$ $term=7^{th}$ $term$

Its general term is $T_{r+1}=\binom{n}{r}a^{n-r}b^r$

Putting the values, we get

$T_{r+1}=\binom{12}{r}\left(\frac{1}{x}\right)^{n-r}\left(\frac{-x^2}{2}\right)^r$

For 7th term put $r=6$ to find the middle term

$T_{6+1}=\binom{12}{6}\left(\frac{1}{x}\right)^{12-6}\left(\frac{-x^2}{2}\right)^6$

$T_7=924\left(\frac{1}{x}\right)^6.\frac{x^{12}}{x^6}=\frac{924x^{12}}{x^6(64)}=\frac{231}{16}x^6$

Which is required.

ii) $\left(\frac{3}{2}x-\frac{1}{3x}\right)^{11}$

Sol: $\left(\frac{3}{2}x-\frac{1}{3x}\right)^{11}$ Middle term=?

Given $\left(\frac{3}{2}x-\frac{1}{3x}\right)^{11}=\left[\frac{3}{2}x+\left(\frac{-1}{3x}\right)\right]^{11}$

Power $n=11$ is even

$\therefore$ Middle term $=\left(\frac{n+1}{2}\right)^{th}$ $term$ and

$\left(\frac{n+3}{2}\right)^{th}$ $term$ and

$\Rightarrow$ Middle term $=\left(\frac{11+1}{2}\right)^{th}$ $term$

And $\left(\frac{11+3}{2}\right)^{th}$ $term$

$\Rightarrow$ Middle term $=6^{th}$ term and 7th term

Now general term is $T_{r+1}=\binom{n}{r}a^{n-r}b^r$

For 6th term put r=5

$T_{5+1}=\binom{11}{5}\left(\frac{3x}{2}\right)^{11-5}\left(\frac{-1}{3x}\right)^5$

$T_6=462\left(\frac{3x}{2}\right)^6\left(\frac{-1}{3^5x^5}\right)$

$T_6=-462\frac{3^6x^6}{2^6}.\frac{-1}{3^5x^5}=\frac{-462\times3\times x}{64}=\frac{-693}{32}x$

Also for 7th term put r=6

$T_{r+1}=\binom{n}{r}a^{n-r}b^r$

$T_{6+1}=\binom{11}{6}\left(\frac{3x}{2}\right)^{11-6}\left(\frac{-1}{3x}\right)^6$

$T_7=462\left(\frac{3x}{2}\right)^5\left(\frac{-1}{3^6x^6}\right)=462.\frac{3^5x^5}{2^5}.\frac{-1}{3^6x^6}$

$T_7=\frac{462}{2^5}.\frac{1}{3.x}=\frac{462}{32\times3\times x}=\frac{77}{16x}$

Which is required.

iii) $\left(2x-\frac{1}{2x}\right)^{2m+1}$

Sol: $\left(2x-\frac{1}{2x}\right)^{2m+1}$ Middle term=?

Given $\left(2x-\frac{1}{2x}\right)^{2m+1}=\left[2x+\left(\frac{-1}{2x}\right)\right]^{2m+1}$

Power $n=2m+1$ is odd

$\therefore$ Middle terms $=\left(\frac{n+1}{2}\right)^{th}$ $term$

And $\left(\frac{n+3}{2}\right)^{th}$ $term$

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$\Rightarrow$ Middle terms

$$=\left(\frac{2m+1+1}{2}\right)^{th} term\ and\ \left(\frac{2m+1+3}{2}\right)^{th} term$$

So, middle terms $=(m+1)^{th}\ term$

And $(m+2)^{th}\ term$

Now $\therefore T_{r+1}=\binom{n}{r}a^{n-r}b^{r}$

For middle term $(n+1)^{th}\ term\ so\ put\ r=m$

$$T_{m+1}=\binom{2m+1}{m}(2x)^{2m+1-m}.\left(\frac{-1}{2x}\right)^{m}$$

$$T_{m+1}=\binom{2m+1}{m}(2x)^{m+1}.\frac{(-1)^{m}}{(2x)^{m}}$$

$$T_{m+1}=\binom{2m+1}{m}.2^{m+1}.x^{m+1}.\frac{(-1)^{m}}{2^{m}x^{m}}$$

$$T_{m+1}=\binom{2m+1}{m}.2^{m+1-m}.x^{m+1-m}.(-1)^{m}$$

$$T_{m+1}=\binom{2m+1}{m}.2^{1}.x^{1}.(-1)^{m}$$

$$T_{m+1}=\frac{(2m+1)!}{m!(2m+1-m)!}2x.(-1)^{m}$$

$$T_{m+1}=\frac{(2m+1)!2x.(-1)^{m}}{m!(m+1)!}$$

Also for $(m+2)^{th}$ term.

Put r = m+1

Putting the values, we get

$$T_{m+2}=\binom{2m+1}{m+1}.(2x)^{2m+1-(m+1)}.\left(\frac{-1}{2x}\right)^{m+1}$$

$$T_{m+2}=\binom{2m+1}{m+1}.(2x)^{2m+1-m-1}.\frac{(-1)^{m+1}}{2^{m+1}x^{m+1}}$$

$$T_{m+2}=\binom{2m+1}{m+1}.(2x)^{m}.\frac{(-1)^{m+1}}{2^{m+1}x^{m+1}}$$

$$T_{m+2}=\frac{(2m+1)!2^{m}.x^{m}.(-1)^{m+1}}{(m+1)!\left[(2m+1)-(m+1)\right]!2^{m+1}.x^{m+1}}$$

$$T_{m+2}=\frac{(2m+1)!(-1)^{m+1}}{(m+1)!(m)!.2^{1}.x^{1}}$$

$$T_{m+2}=\frac{(2m+1)!(-1)^{m+1}}{(m+1)!.m!.2.x}$$

Which is required.

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10. **Show that** $\binom{n}{1}+\binom{n}{2}+\binom{n}{3}+\ldots+\binom{n}{n}=2^{n}-1$

Sol: $\therefore (1+x)^{n}=\binom{n}{0}+\binom{n}{1}x+\binom{n}{2}x^{2}+\binom{n}{3}x^{3}+\ldots+\binom{n}{n}x^{n}$

Put $x=1$, we get

$$(1+1)^{n}=\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\binom{n}{3}+\ldots+\binom{n}{n}$$

$$2^{n}=\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\binom{n}{3}+\ldots+\binom{n}{n-1}+\binom{n}{n}$$

$$2^{n}=1+\binom{n}{1}+\binom{n}{2}+\binom{n}{3}+\ldots+\binom{n}{n-1}+\binom{n}{n}$$

$$2^{n}-1=\binom{n}{1}+\binom{n}{2}+\binom{n}{3}+\ldots+\binom{n}{n-1}+\binom{n}{n}$$

Hence proved $\binom{n}{1}+\binom{n}{2}+\binom{n}{3}+\ldots+\binom{n}{n}=2^{n}-1$

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